

Wavefront .OBJ File Format Guide

Purpose

The `.obj` format is a 3D geometry definition file developed by Wavefront Technologies. It is widely used for representing 3D models in a simple, human-readable text format. It defines the geometry (vertices, faces, texture coordinates, and normals) of a 3D object but does not store full scene information such as lights, cameras, or animation.

History

The OBJ file format was created by **Wavefront Technologies** for its **Advanced Visualizer** animation software in the 1980s. It quickly became a standard in the 3D industry due to its simplicity, portability, and ease of integration with various software applications. Because it is open and text-based, the format has been widely adopted by many 3D modeling and rendering tools, becoming a go-to format for basic geometry exchange between platforms.

File Structure Overview

Prefix	Description
<code>#</code>	Comment line
<code>v</code>	Vertex position (x, y, z)
<code>vt</code>	Texture coordinate (u, v [,w])
<code>vn</code>	Vertex normal (x, y, z)
<code>f</code>	Face (polygon), composed of vertices
<code>o</code>	Object name
<code>g</code>	Group name
<code>usemtl</code>	Material name
<code>mtllib</code>	External material library file (.mtl)
<code>s</code>	Smoothing group

Basic Example

```
# Simple cube example

o Cube
v 0.000000 0.000000 0.000000
v 1.000000 0.000000 0.000000
v 1.000000 1.000000 0.000000
v 0.000000 1.000000 0.000000
v 0.000000 0.000000 1.000000
v 1.000000 0.000000 1.000000
v 1.000000 1.000000 1.000000
v 0.000000 1.000000 1.000000

# Faces (vertices only)
f 1 2 3 4
f 5 6 7 8
f 1 5 8 4
f 2 6 7 3
f 3 7 8 4
f 1 2 6 5
```

Face Definitions and Indexing

Faces (`f`) can reference:

- Vertices only: `f v1 v2 v3`
- Vertices + textures: `f v1/vt1 v2/vt2 v3/vt3`
- Vertices + normals: `f v1//vn1 v2//vn2 v3//vn3`
- All three: `f v1/vt1/vn1 v2/vt2/vn2 v3/vt3/vn3`

Note: Indices are 1-based.

Materials (.mtl) Reference

OBJ files often use a companion `.mtl` file to define material properties:

```
mtllib cube.mtl
usemtl RedPlastic
```

`.mtl` file elements:

- `Ka` – Ambient color
- `Kd` – Diffuse color

- `Ks` – Specular color
 - `Ns` – Shininess
 - `map_Kd` – Texture map image
-

Why It's Useful

- Human-readable and simple to parse
- Supported by most 3D modeling tools (Blender, Maya, Unity, etc.)
- Easy to export/import and convert between applications